

DD: PLM-CFG-06 — Equipment Type Catalog

Developer implementation guide: DD_PLM-CFG-06-Equipment_Type_Catalog-DEV_IMPL_GUIDE-v1.0.md.

1. Purpose

This rewritten DD is the developer-facing version of the module contract
DD_PLM-CFG-06-Equipment_Type_Catalog-v1.0.md.

Verdict: ■ New | **Wave:** 2 | **Group P:** Catalogs (PLM family) | **Version:** v1.0 | **Module:** Product Lifecycle Management — Equipment Catalog The system of record for **all physical equipment types** Wananchi installs at customer premises or holds in stock: routers, ONTs, set-top boxes, splitters, customer-premises cables, wall sockets, mounting kits — every SKU the operator stocks, ships, installs, or returns. Owned at operator scope. Referenced by WO module (work orders bind equipment to install jobs), by OSR-01 Stock Chain Engine (pending, owns stock balances), by FUL-02 STEP-PAYMENT (calculates equipment fees), and by BIL-01 charging (for one-time equipment charges). Retires wo_equipment_ref_stub (DD_WO-01-FRAMEWORK §9.3) — the placeholder schema WO module has been using for equipment-binding at WO finalization is replaced by this DD's structured catalog plus a clean WO→catalog FK.

The goal of this rewrite is to make the DD easier for a mid-level developer to implement without losing the detailed source contract.

2. Implementation Scope

This DD should be implemented as part of the owning SOPHIX capability, not as an isolated service unless the deployment architecture explicitly separates that capability.

Implement this DD with these rules:

- Use the owning module APIs/repositories for authoritative writes.
- Use approved internal APIs/client wrappers when reading another module's data.
- Do not query another module's database tables directly from business flow code.
- Treat worker names as logical Camunda external-task topics, not separate deployments.
- One worker application may handle multiple topics, but metrics, retries, and logs must remain visible per topic.
- Emit success events only after the authoritative state commit succeeds.
- Use outbox publishing for Kafka events.

3. Runtime Metadata

Item	Value
Source file	/Users/macbook/Downloads/Sophix_V3_BSS_DDs_MD_2026-05-27/09_catalogs/DD_PLM-CFG-06-Equipment_Type_Catalog-v1.0.md
Version	v1.0
DD type	module contract
BPMN/process keys	Not explicitly defined
Drools packages	Not explicitly defined

4. Runtime Contracts To Implement

4.1 APIs

- GET /api/equipment-compatibility?operatorCode=WIK&sourceSkuId=WIK-ONT-HG8245-H5
- GET /api/equipment-kinds?operatorCode=WIK&active=true
- GET /api/equipment-skus/{id}/fees
- GET /api/equipment-skus/{sku_id}
- GET /api/equipment-skus/{sku_id}/compatibility

- GET /api/equipment-skus/{sku_id}/fees
- GET /api/equipment-skus?operatorCode=WIK&active=true&kindCode=ONT
- GET /api/equipment-skus?operatorCode=X
- GET /api/equipment-skus?operatorCode=X&kindCode=Y
- PATCH /api/equipment-compatibility/{id}
- PATCH /api/equipment-kinds/{operator}/{kind_code}
- PATCH /api/equipment-skus/{sku_id}
- POST /api/equipment-compatibility
- POST /api/equipment-kinds
- POST /api/equipment-skus
- POST /api/equipment-skus/validate
- POST /api/equipment-skus/{sku_id}/fees

4.2 Tables

- equipment_compatibility
- equipment_fee
- equipment_kind
- equipment_sku

For each table in this DD, implement the schema with:

- a short table comment explaining what the table is for;
- column comments or migration notes explaining each field;
- constraints and indexes from the DD;
- seed data where the DD provides operator-specific sample rows.

4.3 Worker Topics

- No worker topics are explicitly declared in this DD.

Worker-topic guidance:

- A BPMN service task uses the topic.
- A worker application subscribes to the topic.
- The topic handler validates input variables, calls the owning module/API, writes outputs back to Camunda, and records metrics.
- Do not treat the topic string as a shell command or Kubernetes deployment name.

4.4 Events

- EquipmentCompatibilityChanged
- EquipmentFeeChanged
- EquipmentKindCreated
- EquipmentSkuCreated
- EquipmentSkuDeactivated
- EquipmentSkuUpdated
- WorkOrderEquipmentBound

Event guidance:

- Write events through the transactional outbox.

- Include operation id, aggregate id, operator code, actor, and correlation data.
- Consumers should be able to rebuild downstream state without querying workflow internals.

5. Developer Implementation Checklist

1. Add or verify database migrations and seed rows.
2. Add or verify config rows for the operator/module.
3. Implement request/command DTOs and validation.
4. Implement idempotency and in-flight operation checks where the DD requires them.
5. Implement Drools fact mapping and package deployment where rules are used.
6. Implement BPMN process, service tasks, gateways, timers, and message catches where the DD is a flow.
7. Implement worker-topic handlers using owning module APIs.
8. Implement compensation paths and manual recovery paths.
9. Emit success/rejected events through the outbox.
10. Add smoke tests for happy path, validation failure, dependency failure, and retry/idempotency.

6. Infrastructure And AWS Notes

Deploy this DD with the existing SOPHIX platform components unless the owning module requires isolation:

Concern	Recommendation
API/runtime	Run inside the owning module deployment or existing workflow API pods.
Workers	Consolidate low-volume topics into shared worker apps; keep per-topic metrics.
Database	Use the owning module PostgreSQL schema and migration pipeline.
Kafka	Use the foundation Kafka/MSK setup and transactional outbox.
Camunda	Deploy BPMN files through the workflow deployment pipeline.
Drools	Deploy KJAR/rule artifacts through the approved rules pipeline.
Secrets	Use the platform secret manager; on AWS prefer Secrets Manager/Parameter Store with IRSA.
Observability	Alert on stuck operations, failed workers, outbox backlog, and external dependency failures.

7. Original DD Detail Preserved

The following section preserves the detailed source contract for implementation and review.

Verdict: ■ New | **Wave:** 2 | **Group P:** Catalogs (PLM family) | **Version:** v1.0 | **Module:** Product Lifecycle Management — Equipment Catalog

The system of record for **all physical equipment types** Wananchi installs at customer premises or holds in stock: routers, ONTs, set-top boxes, splitters, customer-premises cables, wall sockets, mounting kits — every SKU the operator stocks, ships, installs, or returns. Owned at operator scope. Referenced by WO module (work orders bind equipment to install jobs), by OSR-01 Stock Chain Engine (pending, owns stock balances), by FUL-02 STEP-PAYMENT (calculates equipment fees), and by BIL-01 charging (for one-time equipment charges).

Retires wo_equipment_ref_stub (DD_WO-01-FRAMEWORK §9.3) — the placeholder schema WO module has been using for equipment-binding at WO finalization is replaced by this DD's structured catalog plus a clean WO→catalog FK.

1. What this module is

When a Wananchi installer arrives at Hon. Sarah Wanjiku's address with a router, an ONT, and a wall socket, every piece of equipment they leave behind needs an identity. That identity has several axes: what **kind** of thing is it (router, ONT, splitter, ...), what **brand + model** is it (Huawei HG8245H5, MikroTik hAP ac², ...), what **SKU** does the operator inventory it under (WIK-RTR-HG8245-H5), how much does it cost the customer (deposit + monthly rental), and is it a returnable / consumable /

one-shot. This DD owns all of that.

The catalog is structural and operator-scoped — each operator catalogs their own equipment, sometimes overlapping (Huawei HG8245H5 is sold in KE and UG, with different SKUs and prices). The catalog is read by every module that touches equipment in any way: stock chain, work orders, fulfillment, billing, audit.

What this module does

1. **Catalog equipment SKUs** — operator-scoped, the master identifier (e.g., WIK-RTR-HG8245-H5) that all other modules reference.
2. **Classify equipment by kind** — operator-extensible kind hierarchy: ROUTER, ONT, STB, SPLITTER, WALL_SOCKET, CABLE_PATCH, MOUNTING_KIT, EXTERNAL_ANTENNA, etc. Used by WO module to validate "an install WO must include exactly one ROUTER + exactly one ONT".
3. **Record brand + model** — manufacturer + model number, for ordering, returns, RMA workflows, and customer support context.
4. **Define equipment fees** — one-time install fees, monthly rental fees, deposit (refundable on return). Currency follows operator. References to PLM-CFG-04 discount eligibility codes for promo bundling.
5. **Define equipment ownership semantics** — RETURNABLE (customer must return on disconnect; refundable deposit), CONSUMABLE (no return expected; e.g., a wall socket is left in place), ONE_SHOT (e.g., a cable is consumed at install time; no return), or RENTED (Wananchi-owned, customer keeps custody but ownership stays with operator).
6. **Surface stock signals** (read-only from OSR-01) — does this SKU have stock available right now? (This is a read-through to OSR-01; PLM-CFG-06 does not store stock balances. For v1.0 with OSR-01 pending, PLM-CFG-06 surfaces a placeholder "stock-status-unknown" via OSR-01 stub.)
7. **Record compatibility constraints** — operator-extensible compatibility rules between SKUs (e.g., "this ONT requires this specific splitter type"; "this router is incompatible with this ONT firmware"). v1.0 ships a minimal compatibility table; full constraint resolution lives in WO module.
8. **Track equipment-fee history** — fee changes are versioned (effective_from / effective_to); historical WO + invoice queries resolve to the fee active at the time of the install / billing event.

What this module does NOT do

- **Doesn't track stock balances or movements.** Where the SKU is physically located, how many are in central warehouse vs. contractor van vs. customer premises — all OSR-01 (pending). PLM-CFG-06 is the *definition*; OSR-01 is the *quantity + location*.
- **Doesn't track individual equipment instances.** Each physical router has a serial number; that data lives in OSR-01 (equipment_instance table per OSR-01 plan). PLM-CFG-06 carries only the SKU/type level.
- **Doesn't own provisioning configuration.** Which firmware, which provisioning profile, which RADIUS realm — those belong to provisioning modules (FUL-03 + downstream). PLM-CFG-06 records "this is a Huawei HG8245H5"; what gets pushed to it lives elsewhere.
- **Doesn't manage RMA / return workflows.** RMA, repair, refurbishment, write-off — all OSR-01 + WO module (return work orders). PLM-CFG-06 supplies the SKU metadata they need.
- **Doesn't manage contracts with equipment vendors.** Vendor procurement (POs, vendor SLAs, lead times) is EM-02 (pending — Procurement / Contractor Registry) or a dedicated procurement DD. PLM-CFG-06 just records "brand=Huawei", not "we source it from Huawei East Africa via vendor PO #12345".
- **Doesn't model packaged bundles.** A "starter kit" combining router + ONT + cables is a package modeled in SIP-01 (Package Management) or PLM-CFG-01 (Service Catalog), not here. PLM-CFG-06 records the individual SKUs that bundles reference.

Regional flexibility

Different operators stock different equipment based on network technology, regional vendor relationships, and regulatory constraints:

- **WIK (Kenya)** — primarily Huawei + ZTE for ONTs (Wananchi's GPON network), MikroTik + TP-Link for routers, locally-procured patch cables and wall sockets. Kenya allows direct import; SKU codes follow WIK-<KIND>-<MODEL> convention.

- **WUG (Uganda)** — similar to WIK on technology (also GPON), but local vendor relationships and import constraints differ; some SKUs are UG-specific even for the same physical device. SKU prefix WUG-.
- **WTZ (Tanzania)** — TBD; pending workshop. Likely smaller equipment range. SKU prefix WTZ-.
- **YASSN (Senegal)** — different vendor mix: Senegal market favors Sagemcom + Huawei. SKU prefix YASSN-. French-language display names; deposit amounts in XOF currency.

The schema accommodates all four via:

- Operator-scoped SKU IDs (each operator's catalog is independent)
- TEXT-typed `kind_code` enum-style values (operator-extensible — new kinds land via config, not DDL)
- Currency-aware fee columns (`fee_currency` per row)
- Multi-language display names (`display_name_primary` + `display_name_secondary`) for bilingual operators

Glossary

Term	Meaning
Equipment	Any physical item Wananchi installs, ships, holds in stock, or returns. Includes both customer-side (router, ONT, STB, wall socket, cables at premises) and operator-side (network equipment is OUT of scope — that's RLM/NMS).
SKU	Stock Keeping Unit — the operator's master identifier for one variant of one product. Two routers with the same brand+model but different firmware versions sold for different prices are two SKUs.
Kind	The functional category — what does this device DO. ROUTER, ONT, STB, SPLITTER, WALL_SOCKET, etc. Operator-extensible.
Brand + model	Manufacturer identification (Huawei, ZTE, MikroTik, Sagemcom) + their model designation (HG8245H5, ZXHN H267A, hAP ac², F@st 3686).
Ownership semantics	Whether the customer can keep the equipment, must return it, must pay a deposit, etc. Drives disconnect-flow + dunning-flow behaviors. RETURNABLE / CONSUMABLE / ONE_SHOT / RENTED.
Install fee	One-time charge applied at install for this equipment, in addition to subscription fees. Workshop term often "activation fee" but the catalog separates equipment-specific from service-activation.
Monthly rental	Recurring monthly charge for equipment use; applied via BIL-01 as a recurring line item. For RETURNABLE ownership, monthly rental continues until equipment is returned.
Deposit	Refundable amount charged at install, refunded (via PLM-CFG-05 adjustment) when equipment is returned in working condition. Only applies to RETURNABLE SKUs.
Compatibility	A rule restricting which SKUs can be installed together (e.g., "ONT-X requires SPLITTER-Y"). Read by WO module at WO finalization to flag invalid equipment combinations.

2. Tables overview + relationships

2.1 Tables overview

PLM-CFG-06 owns four tables:

Table	Purpose
<code>equipment_kind</code>	Operator-extensible catalog of equipment kinds (ROUTER, ONT, STB, etc.).
<code>equipment_sku</code>	The master SKU catalog: brand, model, kind, ownership semantics, display names.
<code>equipment_fee</code>	Time-versioned fee schedule per SKU: install fee, monthly rental, deposit, currency.
<code>equipment_compatibility</code>	Rule rows defining which SKUs require / conflict with which other SKUs.

Full DDL in §3.

2.2 Cross-DD anchoring

- **WO module FRAMEWORK §9.3** — currently uses `wo_equipment_ref_stub` for equipment binding at finalization; this DD retires that stub. The retirement plan is in §8.
- **OSR-01 Stock Chain Engine (pending)** — will reference `equipment_sku.sku_id` as the FK for every stock-movement row, every inventory balance, every individual equipment instance.
- **FUL-02 STEP-PAYMENT** — reads `equipment_fee` to add install fee + first month's rental to the order payment calculation.
- **BIL-01 Charging Engine** — applies monthly rental as a recurring line on the customer's account via the `equipment_fee.monthly_rental` value resolved at billing-cycle time (versioned read).
- **PLM-CFG-04 Discount Catalog** — equipment SKUs may be discountable; discount eligibility codes can target specific SKUs (e.g., "free router with annual prepay" promo).
- **PLM-CFG-05 Adjustment Type Catalog** — deposit refunds at disconnect are issued via adjustment codes (ADJ-EQUIPMENT-DEPOSIT-RETURN) defined in PLM-CFG-05; this DD doesn't own the refund mechanism, just the deposit amount.

3. Schemas

3.1 equipment_kind

Operator-extensible catalog of equipment kinds. New kinds added without DDL.

```
CREATE TABLE equipment_kind (
  operator_code TEXT NOT NULL,
  kind_code TEXT NOT NULL,
  display_name_primary TEXT NOT NULL,
  display_name_secondary TEXT NULL,
  description TEXT NULL,
  is_install_required BOOLEAN NOT NULL DEFAULT FALSE,
  install_role TEXT NULL,
  active BOOLEAN NOT NULL DEFAULT TRUE,
  display_order INTEGER NOT NULL DEFAULT 100,
  created_at TIMESTAMP NOT NULL DEFAULT NOW(),
  updated_at TIMESTAMP NOT NULL DEFAULT NOW(),
  updated_by TEXT NOT NULL,
  PRIMARY KEY (operator_code, kind_code)
);

CREATE INDEX idx_eq_kind_active ON equipment_kind (operator_code, active);
```

Sample data:

operator_code	kind_code	display_name_primary	display_name_secondary	is_install_required	install_role
WIK	ROUTER	Router	NULL	true	CPE-PRIMARY
WIK	ONT	Optical Network Terminal	NULL	true	CPE-PRIMARY
WIK	STB	Set-top Box	NULL	false	CPE-SECONDARY
WIK	SPLITTER	Optical Splitter	NULL	false	CABLING
WIK	WALL_SOCKET	Wall Socket / Faceplate	NULL	false	CABLING
WIK	CABLE_PATCH	Patch Cable	NULL	false	CABLING
WIK	MOUNTING_KIT	Mounting Kit	NULL	false	NONE
WUG	ROUTER	Router	NULL	true	CPE-PRIMARY
WUG	ONT	Optical Network Terminal	NULL	true	CPE-PRIMARY
WUG	STB	Set-top Box	NULL	false	CPE-SECONDARY
YASSN	ROUTER	Router	Routeur	true	CPE-PRIMARY
YASSN	ONT	Optical Network Terminal	Terminal Optique	true	CPE-PRIMARY
YASSN	STB	Set-top Box	Décodeur	false	CPE-SECONDARY

`is_install_required` lets WO module validate at finalization that an install WO has equipment from each required kind. KE/UG/SN all require a ROUTER + an ONT on install per current network architecture; STB is optional (only IPTV)

customers); cabling kinds are flexible.

YASSN rows include `display_name_secondary` in French (the rest is operator-language-default).

3.2 equipment_sku

The master SKU catalog.

```
CREATE TABLE equipment_sku (
  sku_id          TEXT PRIMARY KEY,
  operator_code   TEXT NOT NULL,
  kind_code       TEXT NOT NULL,
  brand           TEXT NOT NULL,
  model           TEXT NOT NULL,
  hardware_revision TEXT NULL,
  display_name_primary TEXT NOT NULL,
  display_name_secondary TEXT NULL,
  description      TEXT NULL,
  ownership_semantics TEXT NOT NULL,
  vendor_part_number TEXT NULL,
  manufacturer_country TEXT NULL,
  active          BOOLEAN NOT NULL DEFAULT TRUE,
  is_serialized   BOOLEAN NOT NULL DEFAULT TRUE,
  expected_lifecycle_months INTEGER NULL,
  notes           TEXT NULL,
  created_at      TIMESTAMP NOT NULL DEFAULT NOW(),
  updated_at      TIMESTAMP NOT NULL DEFAULT NOW(),
  updated_by      TEXT NOT NULL,
  FOREIGN KEY (operator_code, kind_code) REFERENCES equipment_kind(operator_code, kind_code)
);

CREATE INDEX idx_sku_operator ON equipment_sku (operator_code, active);
CREATE INDEX idx_sku_kind ON equipment_sku (operator_code, kind_code, active);
CREATE INDEX idx_sku_brand_model ON equipment_sku (brand, model);
```

-- 'WIK-RTR-HG8245-H5' – operator-scoped, human-readable

-- FK composite: (operator_code, kind_code) → equipment_kind

-- 'Huawei' | 'ZTE' | 'MikroTik' | 'Sagemcom' | 'TP-Link'

-- 'HG8245H5' | 'ZXHN H267A' | 'hAP ac²' | 'F@st 3686' | 'Archer C50'

-- 'rev-B' | 'v2' – distinguishes hardware revs same brand+model

-- 'Huawei HG8245H5 (4-port Gigabit ONT)'

-- bilingual

-- 'RETURNABLE' | 'CONSUMABLE' | 'ONE_SHOT' | 'RENTED'

-- vendor's catalog code (for procurement linkage)

-- 'CN' | 'FR' | etc. – for customs/RMA reference

-- TRUE = each instance has a serial number tracked in OSR-01

-- replacement-cycle hint for ops

Sample data:

sku_id	operator_code	kind_code	brand	model	display_name_primary	ownership_semantics	is_serialized	active
WIK-ONT-HG8245-H5	WIK	ONT	Huawei	HG8245H5	Huawei HG8245H5 (4-port Gigabit ONT)	RETURNABLE	true	true
WIK-ONT-ZXHN-H267A	WIK	ONT	ZTE	ZXHN H267A	ZTE ZXHN H267A (Dual-band Wi-Fi 6 ONT)	RETURNABLE	true	true
WIK-RTR-HAP-AC2	WIK	ROUTER	MikroTik	hAP ac²	MikroTik hAP ac² Wireless Router	RETURNABLE	true	true
WIK-RTR-ARCHER-C50	WIK	ROUTER	TP-Link	Archer C50	TP-Link Archer C50 (AC1200 Wireless Router)	RETURNABLE	true	true
WIK-STB-MAG420	WIK	STB	Infomir	MAG420	Infomir MAG420 IPTV Set-top Box	RETURNABLE	true	true
WIK-SPL-1X8-SC	WIK	SPLITTER	Generic	1x8-SC-APC	1x8 SC/APC Optical Splitter	ONE_SHOT	false	true
WIK-WS-FACEPLATE-1G	WIK	WALL_SOCKET	Generic	Faceplate-1G-SC	Single-Gang Faceplate with SC/APC Port	CONSUMABLE	false	true
WIK-CABLE-PATCH-3M-SC	WIK	CABLE_PATCH	Generic	Patch-3m-SC-APC	3m SC/APC Patch Cable	ONE_SHOT	false	true
YASSN-ONT-FAST3686	YASSN	ONT	Sagemcom	F@st 3686	Sagemcom F@st 3686 ONT	RETURNABLE	true	true

sku_id	operator_code	kind_code	brand	model	display_name_primary	ownership_semantics	is_serialized	active
YASSN-RTR-HAP-AC2	YASSN	ROUTER	MikroTik	hAP ac ²	MikroTik hAP ac ² Routeur	RETURNABLE	true	true
WUG-ONT-HG8245-H5	WUG	ONT	Huawei	HG8245H5	Huawei HG8245H5 ONT	RETURNABLE	true	true
WIK-ONT-HG8245-H5-V2	WIK	ONT	Huawei	HG8245H5	Huawei HG8245H5 v2 (refreshed hardware)	RETURNABLE	true	false

Row 6 WIK-SPL-1X8-SC: a passive optical splitter is consumed at install (cabled in, can't reasonably be returned/reused); not serialized.

Row 7 WIK-WS-FACEPLATE-1G: a wall faceplate stays at the customer premises; not returnable, not serialized.

Row 8 WIK-CABLE-PATCH-3M-SC: a patch cable cut to length and terminated; not serialized.

Row 11 WUG-ONT-HG8245-H5: same physical device as row 1 (Huawei HG8245H5) but a separate SKU because the operator + sourcing + warranty differ.

Row 12: a deprecated v2 SKU that's been superseded; active=false means new installs cannot use it but historical records still reference it.

3.3 equipment_fee

Time-versioned per-SKU fee schedule.

```
CREATE TABLE equipment_fee (
  fee_id          TEXT PRIMARY KEY,          -- 'fee_01HZ...' ULID
  sku_id          TEXT NOT NULL REFERENCES equipment_sku(sku_id),
  install_fee_amount NUMERIC(12,2) NOT NULL DEFAULT 0, -- one-time charge at install
  monthly_rental_amount NUMERIC(12,2) NOT NULL DEFAULT 0, -- recurring monthly
  deposit_amount NUMERIC(12,2) NOT NULL DEFAULT 0, -- refundable; only meaningful for RETURNABLE
  fee_currency     TEXT NOT NULL,          -- 'KES' | 'UGX' | 'TZS' | 'XOF'
  tax_inclusive    BOOLEAN NOT NULL DEFAULT TRUE,
  effective_from   DATE NOT NULL,
  effective_to     DATE NULL,              -- NULL = current
  notes           TEXT NULL,
  created_at       TIMESTAMP NOT NULL DEFAULT NOW(),
  created_by       TEXT NOT NULL
);

CREATE INDEX idx_fee_sku          ON equipment_fee (sku_id, effective_from);
CREATE INDEX idx_fee_sku_active  ON equipment_fee (sku_id) WHERE effective_to IS NULL;
CREATE UNIQUE INDEX uq_fee_sku_active
  ON equipment_fee (sku_id) WHERE effective_to IS NULL; -- only one active fee row per SKU
```

The unique partial index enforces that at any given time, each SKU has at most one current fee row. To change a fee: insert a new row with effective_from = today, update the previous row's effective_to = today - 1.

Sample data:

fee_id	sku_id	install_fee_amount	monthly_rental_amount	deposit_amount	fee_currency	effective_from	effective_to
fee_01HZ_WIK_HG8245_V1	WIK-ONT-HG8245-H5	0	0	5000	KES	2023-01-01	2025-06-30
fee_01HZ_WIK_HG8245_V2	WIK-ONT-HG8245-H5	0	0	6500	KES	2025-07-01	NULL
fee_01HZ_WIK_ZXHN_H267A	WIK-ONT-ZXHN-H267A	0	250	7500	KES	2025-01-01	NULL
fee_01HZ_WIK_HAP_AC2	WIK-RTR-HAP-AC2	0	0	8000	KES	2024-04-01	NULL
fee_01HZ_WIK_ARCHER_C50	WIK-RTR-ARCHER-C50	0	0	4500	KES	2024-04-01	NULL
fee_01HZ_WIK_MAG420	WIK-STB-MAG420	2500	350	4500	KES	2024-09-01	NULL
fee_01HZ_WIK_SPL	WIK-SPL-1X8-SC	0	0	0	KES	2023-01-01	NULL

fee_id	sku_id	install_fee_amount	monthly_rental_amount	deposit_amount	fee_currency	effective_from	effective_to
fee_01HZ_WIK_FACEPLATE	WIK-WS-FACEPLATE-1G	800	0	0	KES	2023-01-01	NULL
fee_01HZ_WIK_CABLE_3M	WIK-CABLE-PATCH-3M-SC	0	0	0	KES	2023-01-01	NULL
fee_01HZ_YASSN_FAST3686	YASSN-ONT-FAST3686	0	0	30000	XOF	2024-06-01	NULL
fee_01HZ_YASSN_HAP_AC2	YASSN-RTR-HAP-AC2	5000	0	40000	XOF	2024-06-01	NULL
fee_01HZ_WUG_HG8245_H5	WUG-ONT-HG8245-H5	0	0	200000	UGX	2024-01-01	NULL

Row 1 + 2: KE Huawei ONT deposit raised from 5,000 KES to 6,500 KES on 2025-07-01. Historical WO + invoice queries for the period before 2025-07-01 resolve to row 1; current installs use row 2.

Row 3 KE ZTE ONT carries a monthly rental of 250 KES (this is the higher-end Wi-Fi 6 SKU, rented rather than owned-after-deposit). Note both deposit AND rental — operator can configure either or both.

Row 6 KE Set-top box: install fee + monthly rental + deposit all non-zero (IPTV is a premium tier with explicit fee structure).

Rows 7-9: passive/consumable items priced at zero (cost absorbed in install fee for the service line, not equipment-line; cable cost is in WIK's labor budget).

Row 8 WIK-WS-FACEPLATE-1G: small install fee for the faceplate itself; no deposit (consumable).

3.4 equipment_compatibility

Rule rows defining required-with / incompatible-with relationships between SKUs.

```
CREATE TABLE equipment_compatibility (
  compatibility_id      TEXT PRIMARY KEY,          -- 'compat_01HZ...'
  operator_code         TEXT NOT NULL,
  rule_type             TEXT NOT NULL,            -- 'REQUIRES' | 'INCOMPATIBLE_WITH' | 'RECOMMENDED_WITH'
  source_sku_id         TEXT NOT NULL REFERENCES equipment_sku(sku_id),
  target_sku_id         TEXT NULL,                -- the partner SKU; NULL if rule is kind-level
  target_kind_code      TEXT NULL,                -- alternative: rule targets a kind (e.g., 'any ROUTER')
  severity              TEXT NOT NULL,            -- 'BLOCK' | 'WARN'
  notes                 TEXT NULL,
  effective_from        DATE NOT NULL,
  effective_to          DATE NULL,
  created_at            TIMESTAMP NOT NULL DEFAULT NOW(),
  created_by            TEXT NOT NULL,
  CHECK ((target_sku_id IS NOT NULL) OR (target_kind_code IS NOT NULL))
);

CREATE INDEX idx_compat_operator ON equipment_compatibility (operator_code);
CREATE INDEX idx_compat_source   ON equipment_compatibility (source_sku_id) WHERE effective_to IS NULL;
```

rule_type:

- **REQUIRES** — source SKU **MUST** be paired with target SKU (or target kind) in the same WO. Used for: "this ONT requires the matching splitter".
- **INCOMPATIBLE_WITH** — source SKU cannot be paired with target. Used for firmware mismatches, regulatory restrictions.
- **RECOMMENDED_WITH** — soft hint to dispatcher / installer; not enforced.

severity:

- **BLOCK** — WO module rejects the WO finalization if rule is violated.
- **WARN** — WO module flags the WO for review but allows it.

Sample data:

compatibility_id	operator_code	rule_type	source_sku_id	target_sku_id	target_kind_code	severity	notes
compat_01HZ_HG8245_NEEDS_ROUTER	WIK	REQUIRES	WIK-ONT-HG8245-H5	NULL	ROUTER	BLOCK	Bridge-mode ONT — needs a separate router

compatibility_id	operator_code	rule_type	source_sku_id	target_sku_id	target_kind_code	severity	notes
compat_01HZ_ZXHN_NO_RTR	WIK	INCOMPATIBLE_WITH	WIK-ONT-ZXHN-H267A	NULL	ROUTER	WARN	ONT has integrated Wi-Fi 6; external router redundant but not harmful
compat_01HZ_MAG420_NEEDS_RTR	WIK	REQUIRES	WIK-STB-MAG420	NULL	ROUTER	BLOCK	STB needs Ethernet to a router
compat_01HZ_HG8245_NEEDS_SPLIT	WIK	REQUIRES	WIK-ONT-HG8245-H5	WIK-SPL-1X8-SC	NULL	WARN	Soft recommend the standard splitter; field engineer may sub
compat_01HZ_YASSN_FAST_RTR	YASSN	REQUIRES	YASSN-ONT-FAST3686	NULL	ROUTER	BLOCK	Bridge-mode ONT

Row 1: an install WO that includes WIK-ONT-HG8245-H5 must also include some ROUTER SKU (the rule doesn't say which router — operator-defined; the dispatcher picks based on tier). If the WO is finalized without a ROUTER, WO module rejects it with rule reference compat_01HZ_HG8245_NEEDS_RTR.

Row 2: a soft warn — a WO combining ZXHN ONT (which has Wi-Fi 6 built in) with another router will go through but with a warning logged for the dispatcher to review.

4. Rules

Rule	Description
R-PLM-EQ-1	A sku_id is operator-scoped and human-readable, conventionally <OPERATOR>-<KIND-ABBR>-<MODEL-ABBR> (e.g., WIK-RTR-HG8245-H5). The ID is stable across fee changes, brand corrections, hardware revisions (use a new SKU if hardware diverges substantively).
R-PLM-EQ-2	An equipment_sku row with active=false is hidden from new-install WO equipment-pickers and from FUL-02 STEP-PAYMENT calculations. Historical WO + invoice rows still resolve via the SKU FK; the row remains queryable.
R-PLM-EQ-3	At any moment, each sku_id has exactly zero or one equipment_fee row with effective_to IS NULL (the current fee). The partial UNIQUE index enforces this. Fee changes are append + close-previous in a single transaction.
R-PLM-EQ-4	Historical fee resolution: BIL-01 charging at billing-cycle time uses the fee row where effective_from <= billing_period_start AND (effective_to IS NULL OR effective_to >= billing_period_start). FUL-02 STEP-PAYMENT at order capture uses the fee row active on NOW().
R-PLM-EQ-5	ownership_semantics='RETURNABLE' SKUs MUST have a non-zero deposit in their active fee row (enforced by API validation at fee insert; existing rows are not retroactively validated). CONSUMABLE and ONE_SHOT SKUs MUST have deposit_amount=0. RENTED SKUs MAY have deposit or rental or both.
R-PLM-EQ-6	is_install_required=TRUE on an equipment_kind row means WO module at WO finalization MUST find at least one equipment line with a SKU of that kind. Missing → WO module rejects finalization with reason MISSING_REQUIRED_EQUIPMENT_KIND. (Validation is enforced in WO module per WO-01-FRAMEWORK §9; PLM-CFG-06 declares the requirement.)
R-PLM-EQ-7	Compatibility BLOCK rules are enforced at WO finalization by WO module: if any equipment_compatibility row of rule_type=REQUIRES, severity=BLOCK is not satisfied (target SKU/kind absent), or any rule_type=INCOMPATIBLE_WITH, severity=BLOCK is violated, WO finalization is rejected with EQUIPMENT_COMPATIBILITY_BLOCKED + the compatibility_id. WARN rules log but don't block.
R-PLM-EQ-8	Per-operator kind extensibility: an operator can add a new kind_code value (e.g., EXTERNAL_ANTENNA for some operators, MESH_NODE for others) by inserting an equipment_kind row. Consumers (WO module, OSR-01) must treat unknown kind codes as informational rather than fatal — they may not be able to validate them but they shouldn't reject SKUs they don't recognise.
R-PLM-EQ-9	Operator + kind hierarchy: a WIK SKU references a (operator_code=WIK, kind_code=ROUTER) kind row. Cross-operator references (e.g., a WIK SKU pointing to a WUG kind) are forbidden by the composite FK.

Rule	Description
R-PLM-EQ-10	Deposit refund mechanism: at disconnect, OSR-01 confirms equipment return + condition. Working-condition return triggers a PLM-CFG-05 adjustment (ADJ-EQUIPMENT-DEPOSIT-RETURN) with amount = equipment_fee.deposit_amount resolved at the time the deposit was originally charged (audit trail in BIL-01 charging records). PLM-CFG-06 does not initiate the refund; it provides the historical deposit lookup.
R-PLM-EQ-11	Currency consistency: an equipment_fee.fee_currency must match the operator's default billing currency for the operator owning the SKU. v1.0 does not support multi-currency pricing for the same SKU.
R-PLM-EQ-12	Multi-language display: display_name_primary is the operator's default language (English for WIK/WUG/WTZ; French for YASSN). display_name_secondary is optional bilingual where the operator uses it. WO module BO UI + invoice rendering pick based on user preference language.

5. REST API

5.1 Equipment kind catalog

GET /api/equipment-kinds?operatorCode=WIK&active=true HTTP/1.1
Authorization: Bearer <staff-or-service-jwt>

→ 200 OK

```
{
  "kinds": [
    {
      "operatorCode": "WIK",
      "kindCode": "ROUTER",
      "displayNamePrimary": "Router",
      "displayNameSecondary": null,
      "isInstallRequired": true,
      "installRole": "CPE-PRIMARY",
      "active": true,
      "displayOrder": 10
    },
    {
      "operatorCode": "WIK",
      "kindCode": "ONT",
      "displayNamePrimary": "Optical Network Terminal",
      "isInstallRequired": true,
      "installRole": "CPE-PRIMARY",
      "active": true,
      "displayOrder": 20
    },
    ...
  ],
  "totalCount": 7
}
```

POST /api/equipment-kinds # add a new kind (BO admin only)
PATCH /api/equipment-kinds/{operator}/{kind_code} # update mutable fields

5.2 Equipment SKU catalog

GET /api/equipment-skus?operatorCode=WIK&active=true&kindCode=ONT HTTP/1.1
Authorization: Bearer <staff-or-service-jwt>

→ 200 OK

```
{
  "skus": [
    {
      "skuId": "WIK-ONT-HG8245-H5",
      "operatorCode": "WIK",
      "kindCode": "ONT",
      "brand": "Huawei",
      "model": "HG8245H5",
      "hardwareRevision": null,
      "displayNamePrimary": "Huawei HG8245H5 (4-port Gigabit ONT)",
      "ownershipSemantics": "RETURNABLE",
      "isSerialized": true,
      "active": true,
      "currentFee": {
        "installFeeAmount": 0,
        "monthlyRentalAmount": 0,
        "depositAmount": 6500,
        "feeCurrency": "KES",

```

```

        "effectiveFrom":      "2025-07-01"
      },
      ...
    ]
  }
}

```

The currentFee block is denormalized — GET returns the SKU + its active fee row in one call. This is the high-traffic shape used by FUL-02 STEP-PAYMENT to render the install summary screen.

```

GET /api/equipment-skus/{sku_id}           # detail including all fees, compatibility rules
GET /api/equipment-skus/{sku_id}/fees      # full fee history (active + retired)
GET /api/equipment-skus/{sku_id}/compatibility # compatibility rules where this SKU is source or target
POST /api/equipment-skus                   # create new SKU
PATCH /api/equipment-skus/{sku_id}        # update mutable fields (display name, notes, active flag)

```

5.3 Fee management

```

POST /api/equipment-skus/{sku_id}/fees HTTP/1.1
Authorization: Bearer <bo-admin-jwt>
Content-Type: application/json

```

```

{
  "installFeeAmount": 0,
  "monthlyRentalAmount": 0,
  "depositAmount": 6500,
  "feeCurrency": "KES",
  "taxInclusive": true,
  "effectiveFrom": "2025-07-01",
  "notes": "Deposit raised in line with KES depreciation"
}

→ 201 Created
{
  "feeId": "fee_01HZ_WIK_HG8245_V2",
  "skuId": "WIK-ONT-HG8245-H5",
  "supersededFeeId": "fee_01HZ_WIK_HG8245_V1",
  "supersededEffectiveTo": "2025-06-30"
}

```

Behind the scenes: a transaction that (1) sets the prior current row's effective_to = effectiveFrom - 1 day, (2) inserts the new row. Returns both IDs for audit.

Errors:

- 400 INVALID_FEE_DATES — effectiveFrom <= the previous current row's effective_from.
- 422 CURRENCY_MISMATCH — feeCurrency doesn't match the operator's default.

5.4 Compatibility rules

```

GET /api/equipment-compatibility?operatorCode=WIK&sourceSkuId=WIK-ONT-HG8245-H5 HTTP/1.1
→ 200 OK
{
  "rules": [
    {
      "compatibilityId": "compat_01HZ_HG8245_NEEDS_RTR",
      "ruleType": "REQUIRES",
      "sourceSkuId": "WIK-ONT-HG8245-H5",
      "targetSkuId": null,
      "targetKindCode": "ROUTER",
      "severity": "BLOCK"
    },
    ...
  ]
}

POST /api/equipment-compatibility           # add a new rule
PATCH /api/equipment-compatibility/{id}    # close (set effective_to)

```

5.5 SKU validation for WO finalization

A dedicated endpoint WO module hits at WO finalization to check the equipment-line set against the operator's catalog.

```

POST /api/equipment-skus/validate HTTP/1.1
Authorization: Bearer <wo-service-jwt>
Content-Type: application/json

```

```

{
  "operatorCode": "WIK",
  "woType": "INSTALL",

```

```

    "equipmentLines": [
      { "skuId": "WIK-ONT-HG8245-H5", "qty": 1 },
      { "skuId": "WIK-RTR-HAP-AC2", "qty": 1 },
      { "skuId": "WIK-SPL-1X8-SC", "qty": 1 },
      { "skuId": "WIK-WS-FACEPLATE-1G", "qty": 1 }
    ]
  }
}

→ 200 OK
{
  "valid": true,
  "warnings": [],
  "rulesEvaluated": 4,
  "totalChargesAtTodaysFees": {
    "installFeeAmount": 800,
    "monthlyRentalAmount": 0,
    "depositAmount": 14500,
    "feeCurrency": "KES"
  }
}

```

Or with violations:

```

{
  "valid": false,
  "errors": [
    {
      "code": "MISSING_REQUIRED_KIND",
      "details": "Install WO requires a SKU of kind ROUTER but none provided",
      "missingKind": "ROUTER"
    },
    {
      "code": "EQUIPMENT_COMPATIBILITY_BLOCKED",
      "compatibilityId": "compat_01HZ_HG8245_NEEDS_RTR",
      "details": "WIK-ONT-HG8245-H5 REQUIRES a SKU of kind ROUTER"
    }
  ],
  "warnings": []
}

```

6. Kafka events

PLM-CFG-06 publishes on `sophix.plm.eq.*`.

Event type	Topic	Emitted when	Consumers
EquipmentSkuCreated	<code>sophix.plm.eq.sku-created</code>	New <code>equipment_sku</code> row	OSR-01 (pending), WO module BO UI catalog refresh
EquipmentSkuUpdated	<code>sophix.plm.eq.sku-updated</code>	Mutable fields changed	FUL-02 BO UI catalog refresh
EquipmentSkuDeactivated	<code>sophix.plm.eq.sku-deactivated</code>	<code>active=true→false</code>	OSR-01 (block new stock receipts), FUL-02 (hide from picker)
EquipmentFeeChanged	<code>sophix.plm.eq.fee-changed</code>	New <code>equipment_fee</code> row inserted (with prior row auto-closed)	BIL-01 charging cache invalidation; FUL-02 BO UI fee refresh
EquipmentKindCreated	<code>sophix.plm.eq.kind-created</code>	New <code>equipment_kind</code> row	WO module rule cache invalidation
EquipmentCompatibilityChanged	<code>sophix.plm.eq.compatibility-changed</code>	New compatibility rule or rule closure	WO module rule cache invalidation

6.1 Sample event — EquipmentFeeChanged

```

{
  "eventType": "EquipmentFeeChanged",
  "aggregateId": "WIK-ONT-HG8245-H5",
  "timestamp": "2025-07-01T00:00:00.000Z",
  "payload": {
    "operatorCode": "WIK",
    "skuId": "WIK-ONT-HG8245-H5",
    "kindCode": "ONT",
    "newFeeId": "fee_01HZ_WIK_HG8245_V2",
    "supersededFeeId": "fee_01HZ_WIK_HG8245_V1",
    "oldFee": {
      "installFeeAmount": 0,
      "monthlyRentalAmount": 0,
      "depositAmount": 5000,
      "feeCurrency": "KES",
      "effectiveFrom": "2023-01-01",
    }
  }
}

```

```

    "effectiveTo":          "2025-06-30"
  },
  "newFee": {
    "installFeeAmount":    0,
    "monthlyRentalAmount": 0,
    "depositAmount":       6500,
    "feeCurrency":          "KES",
    "effectiveFrom":        "2025-07-01",
    "effectiveTo":          null
  },
  "changes":               ["depositAmount: 5000 → 6500"],
  "actionRequired":        "FUL-02 STEP-PAYMENT in-progress orders captured before 2025-07-01 retain the old deposit; or"
}
}

```

This event is consumed by BIL-01 to invalidate its charging cache, by FUL-02 BO UI to refresh the install summary screen, and by ops dashboards for audit.

7. Stub retirement

PLM-CFG-06 retires `wo_equipment_ref_stub`, the placeholder schema in DD_WO-01-FRAMEWORK §9.3.

7.1 Stub recap

DD_WO-01-FRAMEWORK §9.3 defines a stub `wo_equipment_ref_stub(wo_id, equipment_ref)` where `equipment_ref` was an opaque string carrying brand/model/qty information without structure. The stub was always known to be temporary; per DD_WO-01-FRAMEWORK's stub-retirement note, when "the future Inventory/Equipment Catalog DD is written" the stub would be replaced with:

- SKUs as first-class entities in PLM-CFG-06 (this DD)
- A WO equipment table referencing `equipment_sku.sku_id` as FK
- Equipment kind collapse into the SKU's kind reference (per row 12 column WO retains only `(wo_id, equipment_ref)` → after retirement, WO has `wo_id + sku_id + qty`)

7.2 Migration steps

1. **Pre-deployment:** PLM-CFG-06 schema deployed; SKU catalog seeded per operator (KE / UG / TZ / SN initial SKUs from operator inventory lists).
2. **WO module schema migration:** new table `wo_equipment(wo_id, sku_id, qty, serial_number_ref)` deployed alongside the old `wo_equipment_ref_stub`. Both coexist during transition.
3. **Backfill:** a one-time job iterates `wo_equipment_ref_stub` rows. For each, parses the opaque `equipment_ref` string (operator-specific parsing logic) and maps `(brand, model)` → `sku_id` by joining with `equipment_sku.brand + .model`. Where the mapping is ambiguous or fails, the row is flagged for manual review; the migration ops team resolves them by hand.
4. **WO module cutover:** WO finalization code switches from `INSERT INTO wo_equipment_ref_stub` to `INSERT INTO wo_equipment`. Pre-finalization SKU validation now calls `POST /api/equipment-skus/validate` (per §5.5) before allowing the WO to commit.
5. **Verification window:** 2 weeks of dual-write — WO module writes to BOTH `wo_equipment_ref_stub` (legacy path) AND `wo_equipment` (new path). A daily reconciliation job compares the two and reports deltas. Production traffic on the new path is monitored.
6. **Removal:** Drop `wo_equipment_ref_stub` table; remove the dual-write code path in WO module; DD_WO-01-FRAMEWORK §9.3 documentation replaced with "retired — see DD_PLM-CFG-06."

7.3 Caller-side changes

In DD_WO-01-FRAMEWORK FINALIZATION step:

7. (Legacy) Write the equipment reference string to `wo_equipment_ref_stub` for downstream provisioning.

becomes:

7. Call `POST /api/equipment-skus/validate` with the equipment lines. If valid, insert one row into `wo_equipment` per line referencing `equipment_sku.sku_id`. Emit `WorkOrderEquipmentBound` event with the structured equipment data (SKU IDs, qtls).

Downstream consumers of WO equipment data (provisioning, billing, OSR-01 stock decrement) now read structured `sku_id` references rather than parsing the opaque string. OSR-01 in particular needs the structured data to track stock movements accurately.

7.4 Historical preservation

`wo_equipment_ref_stub` rows from before migration are preserved during the dual-write window in an archive table `wo_equipment_ref_stub_archive` (created at migration step 6 just before drop). Historical WO queries for pre-migration orders pull from the archive; new queries hit `wo_equipment`.

8. Cross-DD coordination

8.1 Upstream callers

Caller	Endpoint	When
FUL-02 STEP-PAYMENT	GET <code>/api/equipment-skus?operatorCode=X&kindCode=Y</code>	Order capture, when door-to-door agent selects equipment for the install
FUL-02 STEP-PAYMENT	POST <code>/api/equipment-skus/validate</code>	Before finalizing order's equipment list — flags compatibility issues at order time
WO module FRAMEWORK §9 (FINALIZATION)	POST <code>/api/equipment-skus/validate</code>	Before allowing WO finalization — second validation pass (catches changes since order capture)
BIL-01 Charging Engine	GET <code>/api/equipment-skus/{id}/fees</code>	At billing-cycle time, to look up <code>monthly_rental_amount</code> historical fee row for the active billing period
OSR-01 (pending)	GET <code>/api/equipment-skus?operatorCode=X</code>	Stock chain operations: receipts, transfers, audits
BO admin UI	various	Catalog management

8.2 Downstream consumers (Kafka)

Consumer	Events subscribed	Action
FUL-02 BO UI	<code>EquipmentSkuUpdated</code> , <code>EquipmentFeeChanged</code> , <code>EquipmentSkuDeactivated</code>	Refresh equipment-picker cache
WO module	<code>EquipmentSkuDeactivated</code> , <code>EquipmentCompatibilityChanged</code>	Invalidate WO finalization rule cache
OSR-01 (pending)	<code>EquipmentSkuCreated</code> , <code>EquipmentSkuDeactivated</code>	Open / close stock-tracking for the SKU
BIL-01	<code>EquipmentFeeChanged</code>	Invalidate fee cache for monthly charging
Ops dashboards	All <code>sophix.plm.eq.*</code>	Catalog audit + reporting

8.3 Sibling modules

Module	Direction	Contract
PLM-CFG-04 (Discount Catalog)	PLM-CFG-04 → PLM-CFG-06 (read)	Discount eligibility rules may reference specific <code>sku_id</code> values (e.g., "free Huawei HG8245H5 on annual prepay"). Discount evaluation in BIL-01 joins PLM-CFG-04 + PLM-CFG-06 to compute applicable amounts.
PLM-CFG-05 (Adjustment Type Catalog)	PLM-CFG-05 → PLM-CFG-06 (read)	Deposit refund adjustments (ADJ-EQUIPMENT-DEPOSIT-RETURN) read <code>equipment_fee.deposit_amount</code> historically resolved at original install time.
OSR-01 (pending — Stock Chain)	OSR-01 → PLM-CFG-06 (read)	OSR-01's <code>stock_balance</code> , <code>stock_movement</code> , <code>equipment_instance</code> tables all FK to <code>equipment_sku.sku_id</code> . OSR-01 owns the quantities + serials + locations; PLM-CFG-06 owns the definitions.
WO module	WO → PLM-CFG-06 (read + validate)	WO's new <code>wo_equipment</code> table FKs to <code>equipment_sku.sku_id</code> . Validation API gates WO finalization.

Module	Direction	Contract
FUL-02	FUL-02 → PLM-CFG-06 (read + validate)	Order capture reads SKU catalog for the equipment-picker; calls validate API for compatibility check.
BIL-01	BIL-01 → PLM-CFG-06 (read)	Monthly billing reads SKU fees; historical fee resolution per R-PLM-EQ-4.

8.4 Dependencies (build order)

1. **PLM-CFG-04** (Discount Catalog) — already shipped; foundational for fee + discount interactions.
2. **PLM-CFG-05** (Adjustment Type Catalog) — provides the deposit-refund adjustment codes.
3. **Operator currencies** must be configured (BIL-01 or operator config DD).
4. **Equipment kind + SKU + fee seed** — per-operator seed scripts populate the three tables. WIK v1.0 seed includes ~30 SKUs across all kinds; YASSN v1.0 seed includes Sagemcom + MikroTik standard SKUs; WUG / WTZ partial seeds pending full operator inventory data.
5. **WO module migration window** — per §7.2, dual-write to wo_equipment_ref_stub + wo_equipment for 2 weeks; then archive + drop stub.
6. **OSR-01 (pending — does NOT block PLM-CFG-06 v1.0)** — PLM-CFG-06 v1.0 ships and operates without OSR-01. Stock-availability queries from BO UI return "stock-status-unknown" until OSR-01 lands.

9. Workshop sources

Workshop deep-dive sources (**Packages_Services + Inventory**) are present in the project but the PDFs are unreadable for text extraction at time of writing. Equipment functional requirements are referenced implicitly across multiple workshops:

- **Packages & Services workshop** (deep-dive 06) — package definitions reference included equipment (ONT, router); informs the SKU + kind structure.
- **Inventory workshop** (separate deep-dive) — stock chain operations; informs how PLM-CFG-06 catalog must surface to OSR-01.
- **Work Order workshop** (deep-dive 07) — equipment-on-WO finalization, install / shifting / support flows; informs validation rules.

V3 design intent: **provide a structured, operator-extensible equipment catalog** that decouples SKU definitions from stock tracking, fee charging, WO binding, and compatibility validation — each of those concerns lives in its own module. The TZ codebase's current handling of equipment is unverified pending cartography; placeholder: "TBD pending codebase review."

Out of v1.0 scope:

- **Stock balances + movements** — owned by OSR-01 (pending).
- **Individual equipment instance tracking (serial numbers)** — owned by OSR-01.
- **Vendor / procurement records** — owned by EM-02 (pending) or future procurement DD.
- **Firmware / provisioning profile per SKU** — owned by FUL-03 / RLM provisioning modules.
- **RMA workflows** — owned by WO module (return WO types).
- **Bundled equipment kits** — packages live in SIP-01; this DD records only the constituent SKUs.
- **Multi-currency same-SKU pricing** — out; one currency per fee row per operator default.
- **Per-region SKU restrictions** — out; SKU catalog is operator-scoped, not region-scoped. Future enhancement if needed.

End of DD_PLM-CFG-06-Equipment_Type_Catalog-v1.0.